Session: 2018-2019

1. **A. Resolve the following system using the Cramer's Rule:**
   $0.40x_1-0.1x_2-0.2x_3=7.85$
   $0.10x_1+7x_2-0.3x_3=-19.3$
   $0.30x_1-0.2x_2+10x_3=71.4$

Sol^n: Let us write these equations in the form AX=B

$$\begin{bmatrix} .4 & -.1 & -.2 \\ .1 & 7 & -.3 \\ .3 & -.2 & 10 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{bmatrix}$$

$$D=|A|=\begin{vmatrix} .4 & -.1 & -.2 \\ .1 & 7 & -.3 \\ .3 & -.2 & 10 \end{vmatrix}=28.509$$

$$Dx_1=\begin{vmatrix} 7.85 & -.1 & -.2 \\ -19.3 & 7 & -.3 \\ 71.4 & -.2 & 10 \end{vmatrix}=631.059$$

$$Dx_2=\begin{vmatrix} .4 & 7.85 & -.2 \\ .1 & -19.3 & -.3 \\ .3 & 71.4 & 10 \end{vmatrix}=-79.7745$$

$$Dx_3 = \begin{vmatrix} .4 & -.1 & 7.85 \\ .1 & 7 & -19.3 \\ .3 & -.2 & 71.4 \end{vmatrix}=183.027$$

$X_1= Dx_1/D=631.059/28.509\approx22.135$

$X_2= Dx_2/D=-79.7745/28.509\approx-2.79$

$X_3= Dx_3/D=183.027/28.509\approx6.41$

B. use the Gauss-Elimination technique to resolve the following system:
$0.3x_1-0.1x_2-0.2x_3=7.85$
$0.10x_1+7x_2-0.3x_3=-19.3$
$x_1-0.2x_2+10x_3=71.4$

Sol^n: We'll set up the augmented matrix [A|B]:

A|B = | 0.3 -0.1 -0.2 | 7.85 |
| 0.10 7 -0.3 | -19.3 |
| 1 -0.2 10 | 71.4 |

Now, let's perform row operations to solve the system:

Step 1: Make the coefficient of the first row's first element (A[1,1]) equal to 1. Divide the first row by 0.3:

$$A|B = \left|\begin{array}{ccc|c} 1 & -0.3333 & -0.6667 & 26.1667 \\ 0.10 & 7 & -0.3 & -19.3 \\ 1 & -0.2 & 10 & 71.4 \end{array}\right|$$

Step 2: Eliminate the coefficients below A[1,1]. Multiply the first row by -0.10 and add it to the second row. Multiply the first row by -1 and add it to the third row:

$$A|B = \left|\begin{array}{ccc|c} 1 & -0.3333 & -0.6667 & 26.1667 \\ 0 & 7.03333 & -0.2333 & -21.4833 \\ 0 & 0.1333 & 10.6667 & 45.2333 \end{array}\right|$$

Step 3: Make the coefficient of the second row's second element (A[2,2]) equal to 1. Divide the second row by 7.3333:

$$A|B = \left|\begin{array}{ccc|c} 1 & -0.3333 & -0.6667 & 26.1667 \\ 0 & 1 & -0.0318 & -2.9256 \\ 0 & 0.1333 & 10.6667 & 45.2333 \end{array}\right|$$

Step 4: Eliminate the coefficients below A[2,2]. Multiply the second row by -0.1333 and add it to the third row:

$$A|B = \left|\begin{array}{ccc|c} 1 & -0.3333 & -0.6667 & 26.1667 \\ 0 & 1 & -0.0318 & -2.9256 \\ 0 & 0 & 1 & 3.7866 \end{array}\right|$$

Now, the augmented matrix is in row-echelon form, and we can back-substitute to find the solutions:

From the third row, we have:

$1x_3 = 3.7866$

$x_3 = 3.7866$

From the second row, we have:

$1x_2 - 0.0318x_3 = -2.9256$

$x_2 = -2.80$

From the first row, we have:

$1x_1 - 0.3333x_2 - 0.6667x_3 = 26.1667$

$x_1 = 27.71$

So, the correct solution to the system of equations is:

$x_1 \approx 28$

$x_2 \approx -2.80$

$x_3 \approx 3.7866$

2. **A. Apply the Cholesky's process to locate the root of the following system**

X_1-X_2-X_3=2

2X_1+3X_2+5X_3=-3

3X_1+2X_2-3X_3=6.

- **Sol^n:** Rules or technique→ AX=B => LUX=B => LY=B then UX=Y

$$A=\begin{bmatrix}1 & -1 & -1\\ 2 & 3 & 5\\ 3 & 2 & -3\end{bmatrix}, X=\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}, B=\begin{bmatrix}2\\ -3\\ 6\end{bmatrix}$$

A matrix can be written as the product of a lower triangular matrix and an upper triangular matrix

$$A=\begin{bmatrix}1 & -1 & -1\\ 2 & 3 & 5\\ 3 & 2 & -3\end{bmatrix}=\begin{bmatrix}l_{11} & 0 & 0\\ l_{21} & l_{22} & 0\\ l_{31} & l_{32} & l_{33}\end{bmatrix}\begin{bmatrix}u_{11} & u_{12} & u_{13}\\ 0 & u_{22} & u_{23}\\ 0 & 0 & u_{33}\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0 & 0\\ l_{21} & 1 & 0\\ l_{31} & l_{32} & 1\end{bmatrix}\begin{bmatrix}u_{11} & u_{12} & u_{13}\\ 0 & u_{22} & u_{23}\\ 0 & 0 & u_{33}\end{bmatrix}$$

$$=\begin{bmatrix}u_{11} & u_{12} & u_{13}\\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23}\\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+u_{33}\end{bmatrix}$$

Comparing the values,

u_{11}=1, u_{12}=-1, u_{13}=-1, $l_{21}u_{11}$=2, so that l_{21}=2

Similarly, we can find out another unknown variable,

AX=B => LUX=B

$$\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 3 & -1 & 1\end{bmatrix}\begin{bmatrix}1 & 1 & -1\\ 0 & 1 & 7\\ 0 & 0 & 7\end{bmatrix}X=\begin{bmatrix}2\\ -3\\ 6\end{bmatrix}$$

Let, UX=Y => LY=B

$$\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 3 & -1 & 1\end{bmatrix}\begin{bmatrix}y_1\\ y_2\\ y_3\end{bmatrix}=\begin{bmatrix}2\\ -3\\ 6\end{bmatrix}$$

From above matrix we find,

- ✓ y_1=2
- ✓ $2y_1+y_2$=-3
  => y_2=-7
- ✓ $3y_1-y_2+y_3$=6=> y_3=-7

Again, UX=Y

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \\ -7 \end{bmatrix}$$

From above matrix we find,

- ✓ $7x_3=-7$
  $x_3=-1$
- ✓ $x_2+ 7x_3=-7$
  $=> x_2=0$
- ✓ $x_1+ x_2-Y_3=2$

$=> x_1=1$

B.

i. How can we eliminate the error in the Trapezoidal rule by applying the Simpson's rule?

To eliminate error in the Trapezoidal rule using Simpson's rule, follow these steps:

- **Divide the Interval:** Divide the given interval into an even number of subintervals (n) for Simpson's rule. The Trapezoidal rule may use any number of subintervals, but Simpson's rule requires an even number.
- **Apply Trapezoidal Rule:** Use the Trapezoidal rule to approximate the integral over each subinterval individually. This gives you an estimate of the integral using linear approximations.
- **Apply Simpson's Rule:** Now, apply Simpson's rule over the same subintervals. Simpson's rule uses quadratic approximations, which can provide a more accurate estimate.
- **Calculate the Difference:** Subtract the result from Simpson's rule from the result from the Trapezoidal rule for each subinterval.
- **Sum the Differences:** Sum up all the differences calculated in step 4 for each subinterval.

ii. Integrate using Simpson's 3/8 rule.
$f(x)=0.2+25x-200x^2+675x^3-900x^4+400x^5$ from $a=0$ to $b=0.8$

sol^n: A single application of Simpson's 3/8 rule requires four equally spaced points:

- **Points and function values:**
  $f(0)=0.2$

$\frac{a+b}{3} = \frac{0+0.8}{3} = 0.2667$

f(0.2667)=1.432724

0.2667+0.2667=0.5333

f(0.5333)=3.487177

0.5333+0.2667=0.8

f(0.8)=0.232

- **Calculation of approximate integral:**

  using Simpson's 3/8 rule we find,

  $I = 0.8 \text{ X } \frac{0.2+3(1.432724+3.487177)+0.232}{8} = 1.519170$

- **Error Calculation:**

  The true integral value is 1.640533.

  So, E_t = |True Value - Approximate Value|

  = |1.640533 - 1.519170|

  = 0.121363.

  (E_t / True Value) x 100% = (0.121363 / 1.640533) x 100% ≈ 7.4%.

3. A. Solve the following system using the Gauss-Jordan Method:

$$0.7X_1 - 0.1X_2 - 0.2X_3 = 7.85$$
$$0.1X_1 + 7X_2 - 0.3X_3 = -19.3$$
$$X_1 - 0.2X_2 + 10X_3 = 71.4$$

Starting with the augmented matrix:

[0.7 -0.1 -0.2 | 7.85]
[0.1 7 -0.3 | -19.3]
[1 -0.2 10 | 71.4]

Step 1: Divide Row 1 by 0.7:

[1 -0.1429 -0.2857 | 11.2143]
[0.1 7 -0.3 | -19.3]
[1 -0.2 10 | 71.4]

Step 2: Subtract 0.1 times Row 1 from Row 2:

[1 -0.1429 -0.2857 | 11.2143]
[0 7.1429 -0.2857 | -20.1143]
[1 -0.2 10 | 71.4]

Step 3: Subtract Row 1 from Row 3:

[1 -0.1429 -0.2857 | 11.2143]
[0 7.1429 -0.2857 | -20.1143]

[0 0.0571 10.2857 | 60.1857]

Step 4: Divide Row 2 by 7.1429:
[1 -0.1429 -0.2857 | 11.2143]
[0 1 -0.0399 | -2.8157]
[0 0.0571 10.2857 | 60.1857]

Step 5: Subtract 0.0571 times Row 2 from Row 3:
[1 -0.1429 -0.2857 | 11.2143]
[0 1 -0.0399 | -2.8157]
[0 0 10.2857 | 61.7571]

Step 6: Divide Row 3 by 10.2857:
[1 -0.1429 -0.2857 | 11.2143]
[0 1 -0.0399 | -2.8157]
[0 0 1 | 5.9999]

Step 7: Add 0.0399 times Row 3 to Row 2:
[1 -0.1429 -0.2857 | 11.2143]
[0 1 0 | -2.68]
[0 0 1 | 5.9999]

Step 8: Add 0.2857 times Row 3 to Row 1:
[1 -0.1429 0 | 12.498]
[0 1 0 | -2.68]
[0 0 1 | 5.9999]

Step 9: Add 0.1429 times Row 1 to Row 2:
[1 0 0 | 12.498]
[0 1 0 | -2.68]
[0 0 1 | 5.9999]

So, the solution to the system of equations is:
X1 = 12.498
X2 = -2.68
X3 = 5.9999

Rounded to a reasonable number of decimal places, the solution is:
X1 ≈ 12.5
X2 ≈ -2.68
X3 ≈ 6.0

B.

i. **Show the scenarios. In the case of an iteration process, there is convergence and divergence.**

- **Convergence:**
  - **Numerical Approximation:** Imagine you are using the Newton-Raphson method to find the root of a mathematical function.
  - **Gradient Descent (Optimization):** In machine learning, gradient descent is often used to minimize a loss function.
  - **Power Iteration (Eigenvalues):** When finding the largest eigenvalue of a matrix using the power iteration method, the sequence of approximations will often converge to the largest eigenvalue and its corresponding eigenvector.
- **Divergence:**
  - **Numerical Instability:** In some cases, numerical instability can lead to divergence. For instance, when solving a system of linear equations using an iterative method, if the matrix is ill-conditioned, the solution may start to diverge instead of converging to the correct answer.
  - **Learning Rate Too High (Gradient Descent):** In machine learning, if the learning rate in gradient descent is set too high, it can cause the optimization process to diverge.
  - **Chaos Theory:** In some complex systems described by chaos theory, even small changes in initial conditions can lead to divergent behavior

ii. **Write down the algorithm of Bisection Method.**

1. Choose 2 real numbers a & b such that f(a)*f(b)<0.
2. Define root, c=$\frac{a+b}{2}$.
3. Find f(c).
4. If f(a)xf(c)<0
   then set b=c
   else set a=c

return the step-1 until finding the root matched twice.

4 [A] Fit a second-order polynomial to the data of the following table. Also find out standard error $S_{y/x}$.

X_i	Y_i

0	**2.1**
1	**7.7**
2	**13.6**
$\sum x_i = 3$	$\sum y_i = 23.4$

$$(n)a_0 + \left(\sum x_i\right)a_1 + \left(\sum x_i^2\right)a_2 = \sum y_i$$
$$\left(\sum x_i\right)a_0 + \left(\sum x_i^2\right)a_1 + \left(\sum x_i^3\right)a_2 = \sum x_i y_i$$
$$\left(\sum x_i^2\right)a_0 + \left(\sum x_i^3\right)a_1 + \left(\sum x_i^4\right)a_2 = \sum x_i^2 y_i$$

m=2, n=3, $\sum x_i = 3$, $\sum x_i^4 = 17$

$\sum x_i^2 = 5$, $\sum x_i^3 = 9$, $\sum xiyi = 34.9$

$\sum x_i^2 yi = 62.1$, $\sum yi = 23.4$

$\bar{x} = \frac{\sum x_i}{n} = 3/3 = 1$

$\bar{y} = \frac{\sum yi}{n} = 23.4/3 = 7.8$

Therefore, the simultaneous Linear equations are:

$$\begin{bmatrix} 3 & 3 & 5 \\ 3 & 5 & 9 \\ 5 & 9 & 17 \end{bmatrix} \begin{bmatrix} a0 \\ a1 \\ a2 \end{bmatrix} = \begin{bmatrix} 23.4 \\ 34.9 \\ 62.1 \end{bmatrix}$$

Solving these equations through a technique such as Gauss-elimination gives $a_0=7.55$ $a_1=-2.725$ and $a_2=2.875$

$$S_r = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$$

$= 0.14332 + 1.00286 + 1.08158 = 2.22776$

$$s_{y/x} = \sqrt{\frac{S_r}{n - (m + 1)}}$$

$= \sqrt{\frac{2.22776}{3-(2+1)}}$

$= \sqrt{\frac{2.22776}{0}}$

= Something/divided by zero.

=Undefined.

Unfortunately, the results are undefined.

4 B. Derive equation for linear regression and find a_0 and a_1.

The simplest example of a least-squares approximation is fitting a straight line to a set of paired observations: $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$.

The mathematical expression for the straight line is:

$y = a_0 + a_1x + e$---17.1

where a_0 and a_1 are coefficients representing the intercept and the slope, respectively, and e is the error, or residual, between the model and the observations, which can be represented by re-arranging Eq. (17.1) as

$e = y - a_0 - a_1x$

- **Criteria for a "best" fit:** One strategy for fitting a "best" line through the data would be to minimize the sum of the residual errors for all the available data, as in

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - a_0 - a_1x_i) \qquad (17.2)$$

  - ✓ where n = total number of points

  Therefore, another logical criterion might be to minimize the sum of the absolute values of the discrepancies, as in

$$\sum_{i=1}^{n} |e_i| = \sum_{i=1}^{n} |y_i - a_0 - a_1x_i|$$

  A strategy that overcomes the shortcomings of the aforementioned approaches is to minimize the sum of the squares of the residuals between the measured y and the y calculated with the linear model

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_{i,\text{measured}} - y_{i,\text{model}})^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1x_i)^2 \qquad (17.3)$$

- **least-square fit of a straight line:** To determine values for a_0 and a_1, Eq. (17.3) is differentiated with respect to each coefficient:

$$\frac{\partial S_r}{\partial a_0} = -2\sum (y_i - a_0 - a_1x_i)$$

$$\frac{\partial S_r}{\partial a_1} = -2\sum [(y_i - a_0 - a_1x_i)x_i]$$

  Note that we have simplified the summation symbols; unless otherwise indicated, all summations are from i = 1 to n. Setting these derivatives equal to zero will result in a minimum Sr. If this is done, the equations can be expressed as

$$0 = \sum y_i - \sum a_0 - \sum a_1x_i$$

$$0 = \sum y_ix_i - \sum a_0x_i - \sum a_1x_i^2$$

  Now, realizing that $\sum a_0 = na_0$, we can express the equations as a set of two simultaneous linear equations with two unknowns (a_0 and a_1):

$$na_0 + \left(\sum x_i\right) a_1 = \sum y_i \tag{17.4}$$

$$\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 = \sum x_i y_i \tag{17.5}$$

These are called the *normal equations*. They can be solved simultaneously

$$a_1 = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n\Sigma x_i^2 - (\Sigma x_i)^2} \tag{17.6}$$

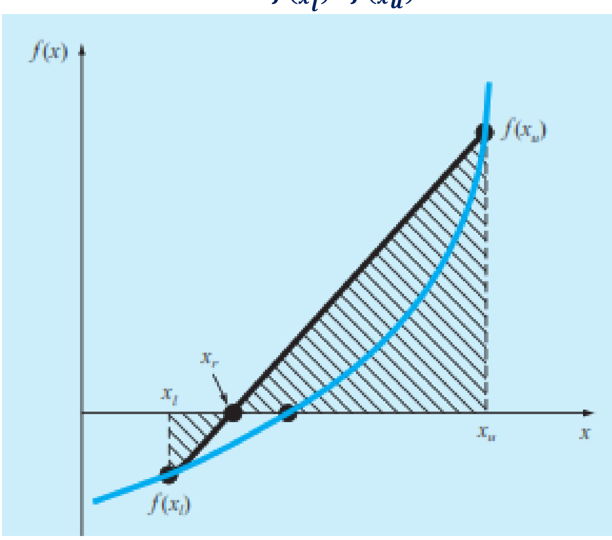
This result can then be used in conjunction with Eq. (17.4) to solve for

$$a_0 = \bar{y} - a_1\bar{x} \tag{17.7}$$

where $\bar{y}$ and $\bar{x}$ are the means of y and x, respectively.

5 A. i. State Weddle's Rule.

ii. show that $x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$



Using triangles, the intersection of the straight line with x-axis can be estimated as:

$$\frac{f(x_l)}{x_r - x_l} = \frac{f(x_u)}{x_r - x_u} \tag{5.6}$$

Cross-multiply Eq. (5.6) to yield

$$f(x_l)(x_r - x_u) = f(x_u)(x_r - x_l)$$

Collect terms and rearrange:

$$x_r\,[f(x_l) - f(x_u)] = x_u f(x_l) - x_l f(x_u)$$

Divide by $f(x_l) - f(x_u)$:

$$x_r = \frac{x_u f(x_l) - x_l f(x_u)}{f(x_l) - f(x_u)} \tag{B5.1.1}$$

This is one form of the method of false position. Note that it allows the computation of the root x_r as a function of the lower and upper guesses x_l and x_u. It can be put in an alternative form by expanding it:

$$x_r = \frac{x_u f(x_l)}{f(x_l) - f(x_u)} - \frac{x_l f(x_u)}{f(x_l) - f(x_u)}$$

then adding and subtracting x_u on the right-hand side:

$$x_r = x_u + \frac{x_u f(x_l)}{f(x_l) - f(x_u)} - x_u - \frac{x_l f(x_u)}{f(x_l) - f(x_u)}$$

Collecting terms yields

$$x_r = x_u + \frac{x_u f(x_u)}{f(x_l) - f(x_u)} - \frac{x_l f(x_u)}{f(x_l) - f(x_u)}$$

or

$$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$

which is the same as Eq. (5.7). We use this form because it involves one less function evaluation and one less multiplication than Eq. (B5.1.1). In addition, it is directly comparable with the secant method which will be discussed in Chap. 6.